Patient 1

Given: BP = 105 Chol = 160

HR = 55

Blood Pressure (Low)

$$\mu(x) = \frac{c-x}{c-b}$$

$$\mu_{low}(105) = \frac{130-105}{130-100}$$

$$= \frac{25}{30} \approx 0.83$$

Cholesterol (low)

$$\mu(x) = \frac{c-x}{c-b}$$

$$\mu_{low}(160) = \frac{200-160}{200-100}$$

$$= 0.40$$

Heart Rate (slow)

$$\mu(x) = \frac{c-x}{c-b}$$

$$\mu_{low}(55) = \frac{80-55}{80-60}$$

$$= 0.83$$

التاريخ : _____

* Rule Strength

Fuzzy AND تستخدم min

Rule Strength = min (0.83, 0.40, 0.83)

= 0.40

بعد تطبيق Centroid Method

CAD ≈ 0.61

ثم القيمة أقل من 1.5 لذلك المريض يصنف Healthy

Patient 2

Given: BP = 120 Chol = 195 HR = 65

Blood Pressure (Low)

$$\mu(120) = \frac{130 - 120}{130 - 100}$$

$$= \frac{10}{30} \approx 0.33$$

Cholesterol (low)

$$\mu(195) = \frac{200 - 195}{200 - 100}$$

$$= \frac{5}{100} \approx 0.05$$

Heart Rate (slow)

$$\mu(65) = \frac{80 - 65}{80 - 50}$$

$$= \frac{15}{30} \approx 0.50$$

Rule strength

$$= \min(0.33, 0.05, 0.50)$$

$$= 0.05$$

CHD $\approx$ 0.73

Not :- Patient 2 is Classified as Healthy

Patient 3

Given : BP = 165 Chol = 186

HR = 95

Blood Pressure (Medium)

$$\mu(165) = \frac{170 - 165}{170 - 120}$$

$$= 0.10$$

Cholesterol (Low)

$$\mu(186) = \frac{200 - 186}{200 - 100}$$

$$= 0.14$$

Heart Rate (Moderate)

$$\mu(95) = \frac{100 - 95}{100 - 70}$$

$$= 0.17$$

Rule strength

$$= \min(0.10, 0.14, 0.17)$$

$$= 0.10$$

CID ≈ 2.63

Note :- Patient 3 is classified as Middle Risk

الجامعية للعلوم التطبيقية
يوم :
تاريخ :

Diagnosis	CHD value	Patient
Healthy	0.61	1
Healthy	0.73	2
Middle Risk	2.63	3

Patient 3 shows higher CHD risk due to higher blood pressure and heart rate values